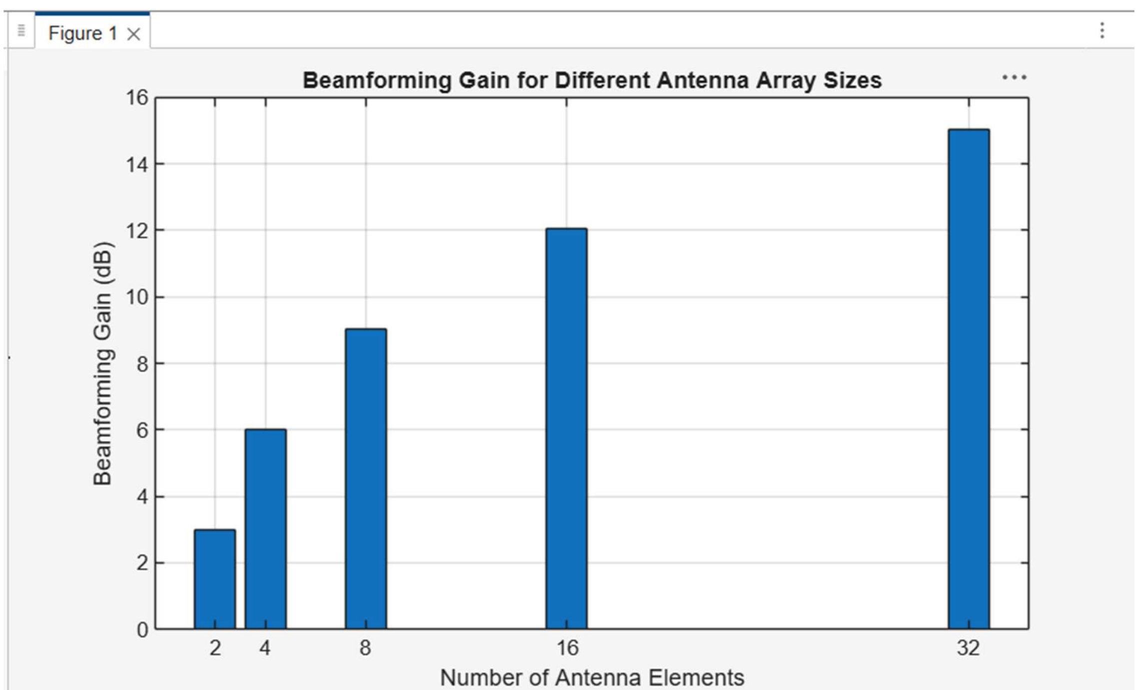
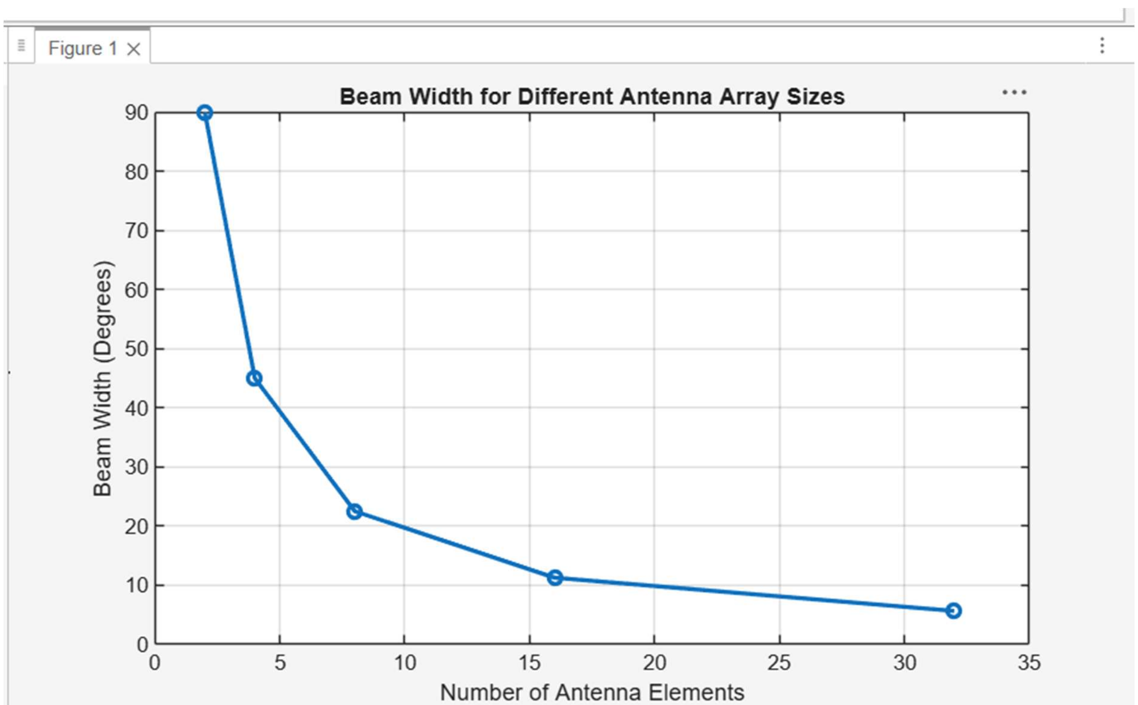
Objective 1

```
clc;
clear;
close all;
antennas = [2 4 8 16 32];
gain = 10*log10(antennas);
figure
bar(antennas, gain)
grid on
% Number of Antenna Elements
% Beamforming Gain (dB)
xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```



Objective 2

```
clc;
clear;
close all;
antennas = [2 4 8 16 32];
% Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625];
figure
% Beam Width (degrees)
plot(antennas, beamwidth,'o-','LineWidth',2)
grid on
xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')
```



Objective 3

```
clc;

clear;

close all;

theta = [-60 -30 0 30 60];

gain = [1 1 1 1 1];

figure

% Main Lobe Direction (degrees)

% Constant normalized gain

polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')
```